

EXPERIMENT 3

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Experiment 3 -- To study and implement advanced SQL queries using Joins, Correlated Query's.

Aim : To study and implement correlated subqueries in SQL by retrieving employees whose salaries are among the top three highest distinct salary values within their respective departments, while correctly handling duplicate salary values using aggregate functions and joins

Question :3 Part A:

You are given with the Employee schema; the task is to find the highest score for each respective department.

For eg: department D1 have the highest score as 5.28. Hence in the output, final updated score for D1 should be 5.28.

Same with the other departments i.e., D2, D3 and D4.

SQL Query:

```
Update Employee e
set scores = t.max_score
FROM (
    Select dept, max(scores) as max_score
    FROM EMPLOYEE
    group by dept
) t
WHERE e.dept = t.dept
-- to view the table
Select * from employee
ORDER BY dept;
```

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	eid [PK] integer	dept character varying (10)	scores numeric (5,2)
1	1	D1	5.28
2	2	D1	5.28
3	3	D1	5.28
4	5	D1	5.28
5	6	D2	8.00
6	4	D2	8.00
7	7	D3	9.00
8	8	D4	10.20

Question: Part B

A company maintains records of products sold by different salespersons. The management wants to prepare a sales performance report by identifying the salesperson(s) who generated the highest overall sales revenue. If multiple salespersons have the same highest total sales amount, all of them must be included in the final report. Return the seller ID(s) satisfying this condition.

SQL Query:

```
SELECT Seller_id
FROM Orders
GROUP BY seller_id
HAVING SUM(amount) = (
    SELECT MAX(am) as max_am
    FROM (
        SELECT seller_id, SUM(amount) as am
        FROM Orders as o
        group by seller_id
    ) t
)
```

	seller_id integer
1	3

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Question: Part C

A company maintains employee salary records for different departments. The management wants to prepare a report containing employees whose salaries fall among the **top three highest distinct salary values** within their respective departments. If multiple employees receive the same qualifying salary, all such employees must be included in the final report. Display the department name, employee name, and salary for every qualifying employee.

SQL Query:

```
SELECT
    d.name AS dept,
    e.name AS employee,
    e.salary
FROM Employee AS e
JOIN Department AS d
    ON e.departmentid = d.id
WHERE (
    SELECT COUNT(DISTINCT e2.salary)
    FROM Employee AS e2
    WHERE e2.departmentid = e.departmentid
    AND e2.salary > e.salary
) < 3
ORDER BY e.salary DESC;
```

	dept character varying (50) 🔒	employee character varying (50) 🔒	salary integer 🔒
1	IT	Max	90000
2	IT	Joe	85000
3	IT	Randy	85000
4	Sales	Henry	80000
5	IT	Will	70000
6	Sales	Sam	60000

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Learning Outcome

After completing this experiment, the learner will be able to:

1. **Understand and implement correlated subqueries** to solve SQL problems involving row-wise comparisons.
2. **Apply aggregate functions** such as COUNT() and DISTINCT to retrieve top-N records within groups.
3. **Use joins and filtering techniques** to combine data from multiple tables and generate meaningful reports.
4. **Analyze and solve complex SQL queries** involving grouped data, ranking logic, and handling duplicate values effectively.